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Directional derivative

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Choose any θ_0 so that $\theta_0(t_0) = p$ and $\theta_0'(t_0) = \mathbf{v}$ and find out $\frac{\mathrm{d}}{\mathrm{d}t}|_{t=t_0} f(\theta_0(t))$

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We define a surface,

$\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and "nice enough" like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

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Recall *Chain rule* (a computational device)

$\frac{\mathrm{d}}{\mathrm{d}t}f(\theta(t)) = \nabla f.\theta'(t)$

$\theta'(t_0) = \ (x'(t_0)\frac{\partial\alpha_1}{\partial x} + \ y'(t_0)\frac{\partial\alpha_1}{\partial y}, \ x'(t_0)\frac{\partial\alpha_2}{\partial x} + y'(t_0)\frac{\partial\alpha_2}{\partial y}, \ x'(t_0)\frac{\partial\alpha_3}{\partial x} + \ y'(t_0)\frac{\partial\alpha_3}{\partial y})$

We define a surface,

$\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and "nice enough" like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the "tangent plane" at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$p = \sigma(x_0, y_0)$

Consider a curve parametrized by $\theta : (a, b) \to \mathbb{R}^3$ which lies on the surface and passes through $p \ = \theta(t_0)$ then $\theta'(t_0)$ is an example of a tangent vector

Since $\theta(t)$ lies on the surface defined by σ,

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$= x'(t_0)\frac{\partial \sigma}{\partial x}|_{x=x_0} + y'(t_0)\frac{\partial \sigma}{\partial y}|_{y=y_0}$

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$= x'(t_0)\frac{\partial \sigma}{\partial x}|_{x=x_0} + y'(t_0)\frac{\partial \sigma}{\partial y}|_{y=y_0}$

Therefore, any tangent vector can be written in terms of the vectors $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

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We can now define the unit normal $\frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial x}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial x} \|}$

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